

BINYAN XU

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Education

The Chinese University of Hong Kong

Sep. 2023 – Aug. 2027 (expected)

Ph.D. Student in Information Engineering; Advisor: Prof. Kehuan Zhang

GPA: 3.86

University of California, Berkeley

Aug. 2021 – Dec. 2021

Exchange Program in EECS

GPA: 4.0

- Selected coursework: Introduction to Embedded Systems (A; Sanjit Seshia), Optimization Models in Engineering (A+; L. El Ghaoui), Dynamic Systems and Feedback (A+; Kameshwar Poolla).

Xi'an Jiaotong University

Sep. 2019 – Jul. 2023

B.Sc. in Automation, Qian Xuesen College; Qian Xuesen Honors Program

GPA: 3.92; Rank: Top 5%

- Entered XJTU from junior high school through the Honors Youth Program, a selective track admitting 100 students from 4,000+ applicants.

Selected Publications and Manuscripts

9 works accepted by or under review at CCF-A conferences, including 7 first/co-first-author works.

- [1] **Binyan Xu***, Xilin Dai*, Kehuan Zhang. Contextual Agentic Memory is a Memo, Not True Memory. Under review at *NeurIPS 2026*. *Equal contribution.
- [2] **Binyan Xu**, Dong Fang, Haitao Li, Kehuan Zhang. From Multi-Agent to Single-Agent: When Is Skill Distillation Beneficial? Under review at *NeurIPS 2026*.
- [3] **Binyan Xu**, Xilin Dai, Fan Yang, Kehuan Zhang. Reviewer Scores Are Not Comparable Across Research Areas in ML Peer Review. Under review at *NeurIPS 2026*.
- [4] **Binyan Xu**, Xilin Dai, Fan Yang, Di Tang, Kehuan Zhang. From Internal Diagnosis to External Auditing: A VLM-Driven Paradigm for Data-Free Online Backdoor Defense. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, PMLR 306, Seoul, South Korea, 2026. **Poster**.
- [5] **Binyan Xu**, Xilin Dai, Fan Yang, Di Tang, Kehuan Zhang. Breaking the Stealth-Potency Trade-off in Clean-Image Backdoors with Generative Trigger Optimization. In *AAAI Conference on Artificial Intelligence (AAAI)*, 2026. **Oral Presentation**.
- [6] **Binyan Xu**, Xilin Dai, Di Tang, Kehuan Zhang. One Surrogate to Fool Them All: Universal, Transferable, and Targeted Adversarial Attacks with CLIP. In *ACM Conference on Computer and Communications Security (CCS)*, 2025. **Oral Presentation**.
- [7] **Binyan Xu**, Fan Yang, Xilin Dai, Di Tang, Kehuan Zhang. CLIP-Guided Backdoor Defense through Entropy-Based Poisoned Dataset Separation. In *ACM International Conference on Multimedia (ACM MM)*, 2025. **Oral Presentation**.
- [8] Fan Yang, **Binyan Xu**, Di Tang, Kehuan Zhang. Beyond Nodes vs. Edges: A Multi-View Fusion Framework for Provenance-Based Intrusion Detection. In *IEEE Symposium on Security and Privacy (S&P)*, 2026.
- [9] Fan Yang, **Binyan Xu**, Di Tang, Kehuan Zhang. Trapping Attacker in Dilemma: Examining Internal Correlations and External Influences of Trigger for Defending GNN Backdoors. Under review at *NeurIPS 2026*.

Research Experience

Agentic Systems: Memory, Skill Distillation, and Evaluation

Jan. 2026 – May 2026

Tencent IEG, Lightspeed Studios, X-Data Research Team; Qingyun Program Internship

Full-time Research Intern

- Led NeurIPS-submitted research on when distilling multi-agent systems into single-agent skills improves performance, introducing *Metric Freedom* as a pre-construction predictor of skill utility.
- Developed adaptive skill-distillation pipelines that match or exceed original multi-agent systems at 1.4–15× lower cost across 4 tasks and 11 datasets.
- Co-led a critical study of contextual agentic memory, arguing that current “memory” modules mostly extend context rather than implement true memory; released code and analysis for reproducibility.
- Built empirical pipelines for agent evaluation, ablation, statistical analysis, prompt optimization, and artifact release.

AI-Aided Aesthetic Graphic Design Generation

Jul. 2022 – Jun. 2023

Microsoft Research Asia, Data, Knowledge and Intelligence; Stars of Tomorrow Internship Program

Full-time Research Intern

- Co-developed an AI-aided aesthetic design system combining autoregressive vision-language layout generation with diffusion-based style and decorative-pattern generation.
- Implemented controllable generation modules to improve visual quality and reduce design-constraint violations.

- Received the *Award of Excellence* in the MSRA Stars of Tomorrow Internship Program for research contribution and full-time participation.

Foundation Model-Aided Neural Network Attacks and Defenses

Sep. 2023 – Jan. 2026

Laboratory for Applied Security Research, CUHK

Ph.D. Research

- Led first-author research on multimodal foundation models for AI security, covering VLM-based external auditing, multimodal-LLM-guided poisoned data separation, CLIP-based adversarial transfer, and generative clean-image backdoors.
- Proposed data-free online backdoor defense using VLMs as external auditors, shifting defense from internal neural diagnosis to externally grounded semantic inspection.
- Designed universal trigger optimization methods that achieve high success rates against real AI services, including Google Image Search, Baidu Image Search, Taobao Image Search, and multimodal models such as GPT-4 and Claude Sonnet.
- Built CLIP-guided attack and poisoned-dataset-separation pipelines; produced four CCF-A first-author publications at ICML, AAAI, CCS, and ACM MM.

Peer Review Fairness and Measurement Design in ML Conferences

Dec. 2025 – Mar. 2026

Laboratory for Applied Security Research, CUHK

Ph.D. Research

- Audited 50,289 ICLR 2021–2026 papers across 219 topics, showing that papers with identical reviewer scores can have up to $8\times$ different acceptance probabilities depending on topic.
- Used regression, likelihood-ratio tests, permutation tests, topic modeling, threshold estimation, and robustness checks.
- Released an interactive analysis website and argued for topic-stratified, score-conditional acceptance rates as a fairness metric for ML conference peer review.

Open-Source and Community Impact

- Open-sourced research code for first-author projects at github.com/binyxu, covering agent evaluation, skill distillation, backdoor attack/defense, adversarial transfer, and peer-review analytics.
- **Research artifacts:** released or prepared reproducible implementations for agent-memory evaluation, adaptive skill distillation, VLM-based external auditing, generative trigger optimization, CLIP-based adversarial transfer, and ICLR topic-level analytics.
- Agent-memory work received broad community discussion after release, including coverage by DAIR.AI on X (26K+ views and 700+ likes/bookmarks), Facebook AI communities, YouTube explainers, Xiaohongshu posts, and Xinzhiyuan coverage.
- Peer-review-fairness analysis was covered by PaperWeekly and discussed on Xiaohongshu, with related posts accumulating 40K+ views, 1,200+ reposts, and 900+ likes.

Teaching and Academic Service

Lead Teaching Assistant, IERG 4998/4999 Final Year Project I/II

2023 – 2026

The Chinese University of Hong Kong

- Guided 40+ undergraduate capstone students across proposal development, technical direction, report feedback, presentation feedback, and grading.
- Coordinated feedback standards for final-year projects and mentored students on research problem formulation, experimental design, technical writing, and presentation delivery.

Academic Service and Conference Presentations

- **Official Reviewer:** ICML (6 papers), ICLR (2), ACM MM (2), AAAI (1).
- **Conference Presentations:** Oral presentations at AAAI, ACM CCS, and ACM MM; poster presentation at ICML.

Selected Awards and Honors

- Finalist Award, Mathematical Contest in Modeling, COMAP, 2021 (Top 2%).
- Qian Xuesen Honors Program Graduate, Xi'an Jiaotong University, 2023.
- Selected into XJTU Honors Youth Program; Top 2.5% among 4,000+ applicants.
- First-class Funding for Berkeley Exchange Program, XJTU, 2021 (Top 5%; approx. RMB 100,000).
- Outstanding Graduate, Qian Xuesen College, Xi'an Jiaotong University, 2023.
- Award of Excellence, Microsoft Research Asia Stars of Tomorrow Internship Program, 2023.

Skills

Research: LLM agents, agent memory and skill consolidation, multimodal foundation models, long-horizon agents, AI security, backdoor/adversarial attacks and defenses.

Programming and Systems: Python, PyTorch, Hugging Face, vLLM, OpenAI API, LangChain, Docker, Linux, Bash, Git, C/C++, HTML/CSS, LaTeX; multi-GPU experimentation up to 16 GPUs.

Languages: Chinese (native), English (fluent; TOEFL 104).